

12.421 / 12.621 Lecture 7

Scattering from nonplanar surfaces

Topics covered

- Radar equation
- Simple analytical models for scattering from rough surfaces
- Geometric properties of rough surfaces
- Scattering from undulating surfaces

Lecture 7

Models for microwave backscattering from rough surfaces

- Motivation for focusing on microwave scattering
 - Reinforce & expand on basic concepts discussed already
 - Useful for SAR observations
 - Sets up discussions of volume scattering and radiometry, which we'll discuss in the next few classes
- Radar equation
 - Let's build more intuition for how we actually measure and quantify scattering
 - This is a heuristic model known as the "radar equation" : Key components
 1. transmit power from antenna
 2. radiated power from the target ("scatterer")
 3. effects of geometry, particularly spreading
 - ↳ Spreading comes from the fact that radiated energy spreads as a sphere
 - ⇒ power density ($\frac{W}{m^2}$) drops off as $\frac{1}{r^2}$ → surface area of sphere with radius r
 - ↳ consistent with the plane wave assumption that we have been making because curvature of the sphere is negligible in the "far-field" where $r \gg \lambda$ (wavelength)

• Radar Eq (cont)

- Let's start from the source
- Consider an antenna w/ transmit power P_t and gain G_t . This has power per unit solid angle at the antenna $= P_t G_t$
- At the scatterer a distance r_t from the antenna, we need to take into account reduction of power density due to spreading. So, the power density at scatterer (magnitude of Poynting vector)

$$S = [P_t G_t] \left(\frac{1}{4\pi r_t^2} \right)$$

$$\frac{\text{Power}}{\text{Area}} = \left[\begin{array}{l} \text{Power transmitted} \\ \text{from antenna} \end{array} \right] \left(\begin{array}{l} \text{spreading} \\ \text{losses} \end{array} \right)$$

- Total power received by the scatterer P_{rs} must account for effective receiving area of scatterer A_{rs}

$$\Rightarrow P_{rs} = S A_{rs}$$

* Note: A_{rs} is not the area illuminated by the incident beam but rather an unknown property of the scatterer

Lecture 7

• Radar Eq (cont.)

- Some power will be absorbed by the scatterer, the rest will be radiated in various directions. Power radiated by the scatterer

$$P_{ts} = P_{is} (1 - f_a)$$

where $f_a \equiv$ fraction of power absorbed by scatterer

- Incident radiation will induce currents (free currents and bound currents due to conductivity and electric displacement $\rightarrow$ i.e., the non-magnetic terms in Ampere's Law)

$\Rightarrow$ can treat the scatterer as an antenna

$$S_r = \frac{P_{ts} G_{ts}}{4\pi r_r^2}$$

where

$G_{ts} \equiv$ gain of scatterer in the direction of the receiver

$r_r \equiv$ distance from scatterer to receiver

- Finally, the power entering receiver is a function of effective area of aperture A_r (not necessarily the physical size of aperture)

$$P_r = S_r A_r$$

Lecture 7

Radar Eq (cont.)

- Putting it all together, we can write an expression for the received power

$$P_r = \underbrace{\left(\frac{P_t G_t A_r}{(4\pi)^2 r_t^2 r_r^2} \right)}_{\text{knowns}} \underbrace{\left[A_{rs} (1 - f_a) G_{ts} \right]}_{\text{unknowns}}$$

- In general, bracketted term is unknown and is a term of interest. Let us define it as

$$\sigma = A_{rs} (1 - f_a) G_{ts} \quad [\text{units: } m^2]$$

and call it the radar (scattering) cross section, as it has units of m^2 (note: in the class, I have tried to use variables in a way that is consistent with usage in RS, and here we have encountered one of the unfortunate inconsistencies in conventional variables, with σ now being used to denote something other than electrical conductivity)

- More commonly, we normalize by the nominal area of illumination to get the normalized radar cross section (sigma-naught)

$$\sigma^0 = \left\langle \frac{\sigma_i}{A_{i,i}} \right\rangle \quad \text{where} \quad \sigma_i \equiv \text{radar cross section for individual scatterer}$$

and $\langle \cdot \rangle$ denotes spatial average

Lecture 7

• Radar Eq. (cont.)

- Up to this point, we have ignored wave polarization for simplicity. We can account for polarization by noting that the transmit and received power ~~are~~ polarization dependent

$P_p^r \equiv p$ -polarized received power (horizontal or vertical)

$P_g^t \equiv g$ -polarized transmit power (h or v)

$$P_p^r = \frac{\lambda^2}{(4\pi)^3} \int_{\text{area illuminated}} \frac{P_g^t G^2}{r^4} \sigma_{pg}^0 dA$$

where we have assumed

$$G_t = G_r = G \quad \text{and} \quad r_t = r_r = r$$

and applied the identity

$$A_r = \frac{\lambda^2 G_r}{4\pi}$$

- For intuition, we can note that σ_{pg}^0 is $\xrightarrow{\text{scattered electric field}}$ scattered electric field

$$\sigma_{pg}^0 = \frac{4\pi r^2 |E_p^s|^2}{A |E_g^i|^2} = \frac{4\pi r^2}{A} \Gamma_{pg}$$

where Γ_{pg} is the reflectance, a generalization of the reflectance we derived before for planar interfaces

Lecture 7

- Scattering from nonplanar surfaces

- Now that we have a basic sense of how we often quantify the scattering of EM waves, let's consider some simple models for surfaces
- We'll focus on simple analytical models
- Simple analytical models only exist for a few special cases that occur when the spatial dimensions of the surface roughness are much smaller or much larger than the EM wavelength
 - ↳ For the intermediate case, when roughness is EM wavelength, very complex models are necessary. We won't go over these in class because they offer little additional insight. But one such model, known as the Integral Equation Model (IEM) is discussed in Ulaby et al (2014) Chapter 10 and in the primary literature Fung (1992, 2002) for those who are interested
- Basic concepts of surface scattering
 - For surface scattering, we are concerned with
 - Total power reflected by surface (reflectivity)
 - direction of scattering

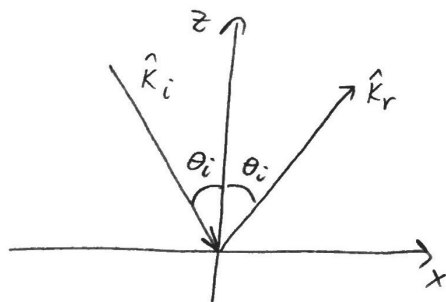
Lecture 7

• Scattering from rough surfs (cont.)

- Basic concepts (cont.)

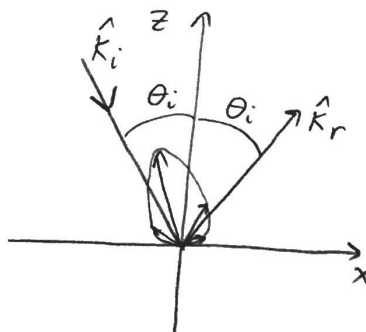
- Scattering from an area is simply (re) radiation from the area $\Rightarrow$ many concepts discussed here are adopted from antenna theory (which we will not discuss in detail but those interested in learning more can consult Ulaby and Long (2014), ch 3)
- Key to understanding the direction of scatter is the surface roughness

perfectly smooth



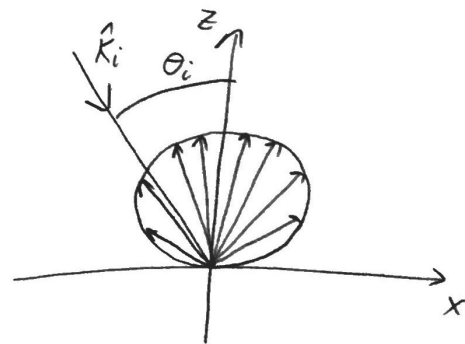
"Specular" or coherent scattering

- Reflected wave has uniform phase front $\Rightarrow$ coherent
- All reflected power follows single ray path



- Combination of coherent and incoherent scatter
- Natural surfaces typically fall in this category

perfectly rough



"diffuse" or incoherent scatter

- A.K.a. Lambertian reflection
- Radiance (apparent brightness) is same in all directions

Lecture 7

• Rough surfaces (cont.)

- How do we define surface roughness?

2 length scales

1. RMS height h

2. correlation length l

- Definitions:

• Let us define local surface height as a 2D

function $z(x, y)$

• Assume z has a Gaussian distribution w/ zero mean

• Therefore, RMS height

$$h^2 = \langle z^2(x, y) \rangle$$

• But we need more info, specifically how height at one point relates to height at another point

$\Rightarrow$ auto-correlation function

$$\rho(r) = \frac{\langle z(x, y) z(x', y') \rangle}{h^2}$$

where $r = \sqrt{(x-x')^2 + (y-y')^2}$ is the distance between points

• 2 correlation functions are commonly assumed

- Gaussian: $\rho(r) = e^{-r^2/l^2}$

- Exponential: $\rho(r) = e^{-|r|/l}$

$\Rightarrow$ makes clear that l is distance over which correlation decreases by $1/e$

Lecture 7

• Rough Surf (cont.)

- More common to represent surface in terms of roughness spectrum

$$W(k_x, k_y) = \frac{1}{2\pi} \iint_{-\infty}^{\infty} \rho(u, v) \exp\{-j(k_x u + k_y v)\} du dv$$

where

$$k_x = -k \sin \theta_s \cos \phi_s \quad \text{and} \quad k_y = k \sin \theta_s \sin \phi_s$$

where

$\theta_s \equiv$ scattering angle

$\phi_s \equiv$ scattering azimuth (angle between plane of incidence and propagation vector of the scattered wave, measured in the horizontal plane)

EX., for the case where a signal is transmitted and received from same antenna, $\theta_s = \theta_i$ and $\phi_s = \pi$

For common correlation functions, we have

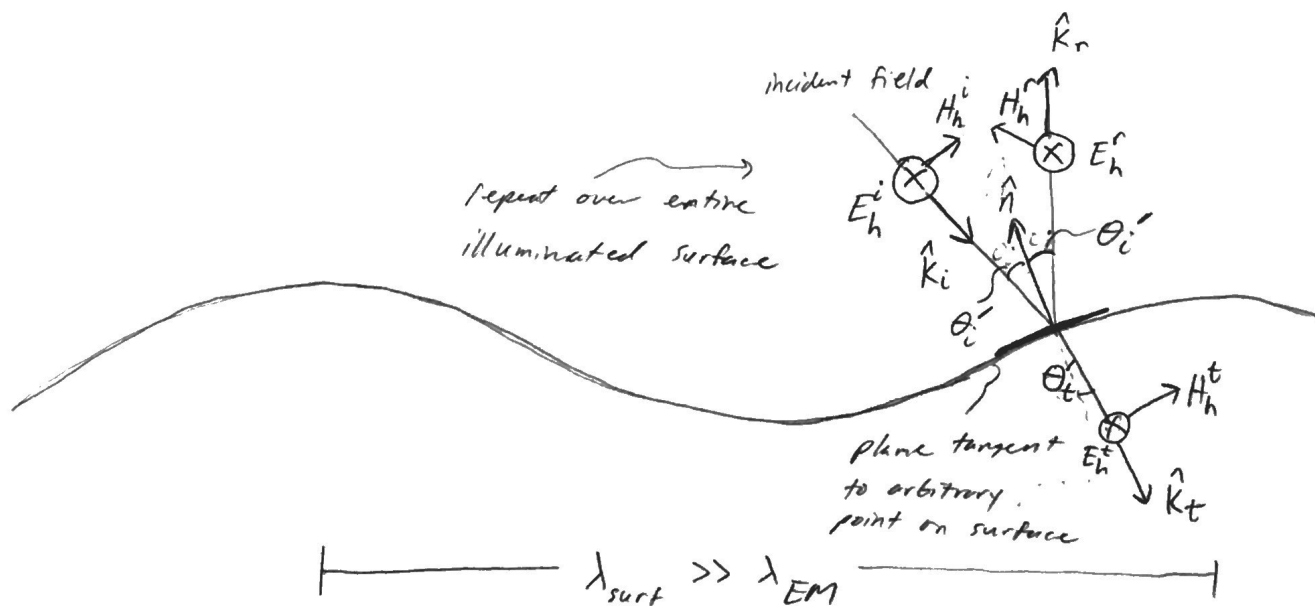
$$\text{- Gaussian: } W(k_x, k_y) = \frac{l^2}{2} \exp\left\{-\frac{(k_x^2 + k_y^2) l^2}{4}\right\}$$

$$\text{- Exponential: } W(k_x, k_y) = \frac{l^2}{[1 + (k_x^2 + k_y^2) l^2]^{3/2}}$$

Lecture 7

• Scattering from an undulating surface

- Commonly called Kirchhoff or physical optics model
- Applicable to surfaces w/ gentle undulations whose wavelength $\gg$ EM wavelength. Allows us to
 1. Approximate surface as a series of infinite planes that are tangent to the surface (assumption of infinite plane allows us to neglect boundary effects)
 2. Use concepts of reflection and transmission at a planar boundary as discussed in previous lectures
- Details get messy and we won't go through them in this course (I'm happy to provide references for those who are interested)
- Basic concepts



Lecture 7

• Undulating surface (cont.)

- Can calculate scattered and transmitted fields, just as we did for the case of planar interface (because all we've done is approximated the undulating surface as a bunch of planar surfaces)

$\Rightarrow$ Fresnel reflection and transmission coefficients will play a role

- To "keep things simple" and the course moving forward, we'll only consider the question of how much power is scattered back toward the source (as is the general case in "active" remote sensing)

- Again, skipping the details and simply noting that to do so, we need to consider the entire scattered field $\Rightarrow$ integrate over the surface
 $\rightarrow$ done by application of Green's Theorem

- To get an analytical solution, it is necessary to make an assumption about the RMS height of the surface h

1. Limit of small h so that surface slopes are shallow:

- allows for a reduced model for backscatter commonly called "scalar approximation" (pretty terrible name, but that's what we have)

Lecture 7

- "scalar approximation" expression for backscatter is pretty gnarly and we won't repeat it here $\rightarrow$ see Ulaby et al. (1986) Ch 12 for expression and details of derivation
- Key point is that shallow slopes allow for both coherent (specular) and incoherent (diffuse) scatter $\rightarrow$ in general, backscattered power will increase with roughness height and with decreasing wavelength (i.e., steepening surface slope)

2. Limit of large h ($h \equiv$ RMS roughness height)

- Allows us to assume the gradient of phase is negligible ("stationary phase approximation")
- It can be shown in this case that

$$\sigma_{pp}^0 = \frac{|P_p(0)|^2}{4m^2 \cos^4 \theta_i} \exp \left\{ -\frac{\tan^2 \theta_i}{4m^2} \right\}$$

$p = h$ or v polarization

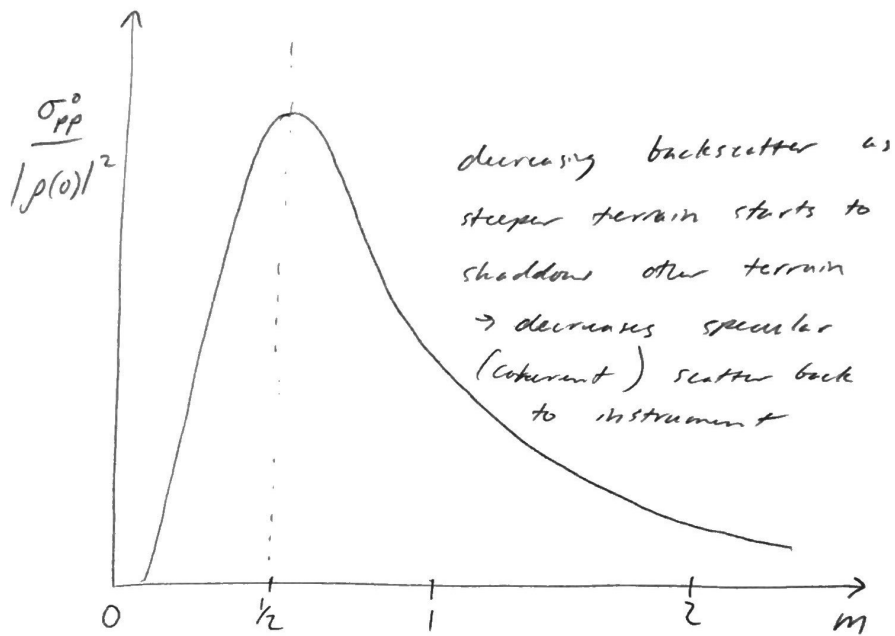
$m = \frac{h}{\lambda} \equiv$ characteristic surface slope

$P_p(0) = p(0) \equiv$ Fresnel reflection coefficient at normal incidence

(note $\sigma_{hv}^0 = \sigma_{vh}^0 = 0$ in this model $\Rightarrow$ no depolarization)

Lecture 7

- Plotting this expression for $\theta_i = 45^\circ$ and a range of slopes



~~~~~  
increasing backscatter  
as slopes steepen  
and increase  
specular scatter  
toward instrument  
receiver

- applicability of model to natural terrain is limited due to restrictions of the assumptions, but the intuition we build from the approach (simplifying assumptions) and results (backscatter power is amplified for certain terrain) is valuable for more realistic scattering models